

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12413882
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Wang; Xinhong

Personal audio system and application thereof

Abstract

A personal audio system and an application thereof are disclosed. The personal audio system includes a housing, in which a right front loudspeaker, a left front loudspeaker and a controller are arranged. A right rear conduction tube is arranged on the right front loudspeaker. An opening at one end of the right rear conduction tube is tightly connected and sealed to one side of the right front loudspeaker, and an opening at the other end of the right rear conduction tube is away from a right ear. The right front loudspeaker and the left front loudspeaker are arranged symmetrically.

Inventors:	Wang; Xinhong (Sichuan, CN)
Applicant:	SHENZHEN LONGWEI TECHNOLOGY CO., LTD (Shenzhen, CN)
Family ID:	75695851
Assignee:	SHENZHEN LONGWEI TECHNOLOGY CO., LTD (Shenzhen, CN)
Appl. No.:	18/258699
Filed (or PCT Filed):	December 11, 2021
PCT No.:	PCT/CN2021/137274
PCT Pub. No.:	WO2022/135191
PCT Pub. Date:	June 30, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240048882 A1	Feb. 08, 2024

Foreign Application Priority Data

CN	202011569907.1	Dec. 26, 2020
----	----------------	---------------

Publication Classification

Int. Cl.: H04R1/02 (20060101)

U.S. Cl.:

CPC H04R1/025 (20130101); H04R2201/029 (20130101); H04R2499/13 (20130101)

Field of Classification Search

CPC: H04R (1/025); H04R (2201/029); H04R (2499/13)

USPC: 381/386

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4038499	12/1976	Yeaple	N/A	N/A
10555071	12/2019	Wakeland	N/A	G02C 11/10
2019/0118688	12/2018	Fujikake	N/A	N/A
2020/0374631	12/2019	Hayashi	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201345722	12/2008	CN	N/A
108407685	12/2017	CN	N/A
112770221	12/2020	CN	N/A
H07264689	12/1994	JP	N/A
H09331592	12/1996	JP	N/A
2002281579	12/2001	JP	N/A
9806232	12/1997	WO	N/A
2020125940	12/2019	WO	N/A
2020125941	12/2019	WO	N/A

OTHER PUBLICATIONS

International Written Opinion of corresponding PCT application, PCT/CN2021/137274, dated Mar. 8, 2022, 4 pages. cited by applicant

International Search Report of corresponding PCT application, PCT/CN2021/137274, dated Mar. 8, 2022, 5 pages. cited by applicant

Search Report from corresponding European Application No. 21909188.1, mailed Jun. 7, 2024, 5 pages. cited by applicant

Search Report from corresponding Japanese Application No. 2023-561416, mailed Jul. 29, 2024, 8 pages with translation. cited by applicant

Primary Examiner: Nguyen; Sean H

Attorney, Agent or Firm: Tucker Ellis LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage filing under 35 U.S.C. § 371 of international application number PCT/CN2021/137274, filed Dec. 12, 2021, which claims priority to Chinese patent application No. 202011569907.1 filed Dec. 26, 2020. The contents of these applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The disclosure relates to the field of audio device for replaying music, in particular to a personal audio system which can be used in a headrest or a neck pillow in various environments such as automobiles, airplanes and high-speed rails.

BACKGROUND

(3) Most loudspeakers of modern automobile audios are mounted in door panels or center consoles, but such an installation method has the following defects:

(4) 1. Sound field is not centered: Since distances between a seat and door panels on two sides of the seat are different, the sounds from a left loudspeaker and a right loudspeaker will be transmitted to human ears at different times with different intensities, so that the sound field cannot be centered.

(5) 2. Sound field is difficult to be distinguished: With the development of the society, more and more people desire to hear their favorite music on vehicles, or nearby people do not want to be disturbed when listening to music, or people do not want others beside them to overhear the conversation of callers when communicating by means of automobile audio device. Current automobile audio device cannot meet such a requirement.

(6) 3. The cost is high: At present, mid-to-high-end automobile audio device on the market has a large number of loudspeakers. For example, Cadillac CT6 has been developed to have 34 loudspeakers, and a new generation product has more than 48 loudspeakers.

(7) Although a headset can overcome most of the above defects, the headset needs to be worn on the head of a user, bringing inconvenience to a certain degree. Also, the headset blocks the ears of the user, which brings inconvenience to the user communicating with others beside the user.

(8) Loudspeakers are mounted on some seats, and can play music for personal appreciation. However, such design uses the design thought of a loudspeaker box, and a loudspeaker and a box body which can be mounted on the seat are small in size, resulting in a very poor bass effect, so that the requirement for high-quality music replay is not met.

SUMMARY

(9) The disclosure aims to overcome the defects in the prior art, and provides a personal audio system which can be mounted on a headrest of a seat or a neck pillow against which the shoulders of a user lean.

(10) Due to the fact that sound waves radiated from the front side and the back side of a loudspeaker are the same in amplitude and opposite in phase, under the condition of no blocking, the sound waves radiated from the front side and the back side offset each other, resulting in poor bass which means acoustic short circuiting. In order to prevent the acoustic short circuiting and improve the bass, it is necessary to manage to prevent the sound waves from the front side and the back side of the loudspeaker from being radiated to the same space. A cavity of a loudspeaker box is used for limiting sound waves from the back side of the loudspeaker in an independent space, but the added cavity can cause the lowest resonant frequency of the loudspeaker to increase, resulting in poor bass. According to the disclosure, a novel design thought is used to prevent or weaken the influence of acoustic short circuiting.

(11) The sound wave has a characteristic that each time a propagation distance in a free field space

is doubled, the amplitude will be attenuated by half (6 dB). Therefore, if the loudspeaker and a listening position can be kept relatively stable, the sound waves from one side of the loudspeaker can be transmitted to a farther position by means of a conduction tube, and the other side of the loudspeaker is adjacent to the ear. In this way, when the sound waves at a distal end in the free field space are propagated back to the ears, the amplitude is greatly lower than that of the sound waves at a proximal end, thereby obtaining a good bass effect. Particularly, when a propagation distance difference in the free field space is not less than 8 times, the offset amplitude is less than 1 dB, and it can be equivalently considered that no mutual offset occurs. Moreover, by adding a tube to the loudspeaker, the lowest resonant frequency of the loudspeaker can be reduced, and the bass is improved.

(12) However, for people at a distance (the distance is greater than 500 mm), propagation distances of the sound waves from the front side and the back side of the loudspeaker are almost the same, the sound waves are still similar in amplitude and opposite in phase and can offset each other, and thus weakening interference with the people at a distance. In this manner, high-quality private sound replay can be achieved.

(13) Since this equipment does not need to cover the ears of a user, it does not interfere with the communication between a user and other persons or the user can hear external sound needed. In addition, no matter whether it is mounted on a seat or worn on the shoulder, the burden on the user is greatly reduced.

(14) The disclosure is implemented by using the following technical solution.

(15) A personal audio system is provided comprising a housing **10** in which a right front loudspeaker **31**, a left front loudspeaker **32** and a controller **50** are all arranged, wherein a right rear conduction tube **41** is arranged on the right front loudspeaker **31**, an opening at one end of the right rear conduction tube **41** is tightly connected and sealed to one side of the right front loudspeaker **31**, and an opening at the other end of the right rear conduction tube **41** is away from a right ear of a user; a left rear conduction tube **42** is arranged on the left front loudspeaker **32**, an opening at one end of the left rear conduction tube **42** is tightly connected and sealed to one side of the left front loudspeaker **32**, and an opening at the other end of the left rear conduction tube **42** is away from a left ear of a user.

(16) In the personal audio system as described above, a right front conduction tube **43** is arranged on the right front loudspeaker **31**, an opening at one end of the right front conduction tube **43** is tightly connected and sealed to the other side of the right front loudspeaker **31**, and an opening at the other end of the right front conduction tube **43** is close to the right ear of the user; a left front conduction tube **44** is arranged on the left front loudspeaker **32**, an opening at one end of the left front conduction tube **44** is tightly connected and sealed to one side of the left front loudspeaker **32**, and the opening at the other end of the left front conduction tube **44** is close to the left ear of the user.

(17) In the personal audio system as described above, the right front loudspeaker **31** and the left front loudspeaker **32** are the same in structure and are symmetrically arranged, the left front conduction tube **44** and the right front conduction tube **43** are the same in structure and are symmetrically arranged, a distance a between the right front conduction tube **43** and an ear canal opening ranges from 30 mm to 200 mm, and a distance by which a sound wave is radiated by air from a distal end of the right rear conduction tube **41** to the right ear is represented by b , wherein $b/a > 2$.

(18) In the personal audio system as described above, $b/a > 8$, an error between sectional areas of the right front conduction tube **43** and the right rear conduction tube **41** does not exceed 10%, and an error between lengths of the right front conduction tube **43** and the right rear conduction tube **41** does not exceed 10%.

(19) In the personal audio system as described above, the right front loudspeaker **31** and the left front loudspeaker **32** are the same in structure and are symmetrically arranged, the right rear

conduction tube **41** and the left rear conduction tube **42** are the same in structure and are symmetrically arranged, and radiation axes of the right front loudspeaker **31** and the left rear conduction tube **42** respectively point to two ears of the user; an included angle θ formed by the two axes ranges from 30° to 170° , a distance a by which a sound wave is radiated from a front side of the right front loudspeaker **31** to the right ear pointed by the right front loudspeaker **31** ranges from 30 mm to 200 mm, a distance by which a sound wave is radiated by air from a distal end of the right rear conduction tube **41** to the right ear is represented by b , wherein $b/a > 2$.

(20) In the personal audio system as described above, $b/a > 8$.

(21) In the personal audio system as described above, the right front loudspeaker **31** and the left front loudspeaker **32** are the same in structure and are symmetrically arranged, a right rear loudspeaker **33** and a left rear loudspeaker **34** are symmetrically arranged in the housing **10**, the right rear loudspeaker **33** is reversely arranged behind the right front loudspeaker **31**, and the left rear loudspeaker **34** is reversely arranged behind the left front loudspeaker **32**.

(22) An application of the personal audio system as described above is provided, the personal audio system is mounted in a headrest of a seat.

(23) An application of the personal audio system as described above is provided, the personal audio system is mounted in a neck pillow.

(24) Compared with the prior art, the disclosure has the following advantages.

(25) 1. The personal audio system of the disclosure does not need to be in contact with ears, so that the burden on the user is not increased, and the personal audio system is comfortable to use.

(26) 2. The personal audio system of the disclosure is good in tone quality effect and rich in bass. If the loudspeaker is additionally provided with a cavity, the resonant frequency will increase, resulting in poor bass. However, if the loudspeaker is additionally provided with a conduction tube, the resonant frequency will decrease, resulting in a richer bass.

(27) 3. The personal audio system of the disclosure is good in lateral symmetry and natural in sound field.

(28) 4. The personal audio system of the disclosure can be used with good privacy, and does not disturb others and also enhances the privacy of communication.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic structural diagram of embodiment one of the disclosure;

(2) FIG. 2 is a schematic structural diagram of embodiment two of the disclosure;

(3) FIG. 3 is a schematic structural diagram of embodiment three of the disclosure;

(4) FIG. 4 is a schematic structural diagram of embodiment four of the disclosure.

(5) In the figures: **10**—Housing, **31**—Right front loudspeaker, **32**—Left front loudspeaker, **50**—Controller, **41**—Right rear conduction tube, **43**—Right front conduction tube, **33**—Right rear loudspeaker, **34**—Left rear loudspeaker, **44**—Left front conduction tube, and **42**—Left rear conduction tube.

DETAILED DESCRIPTION

(6) For ease of understanding by a person of ordinary skill in art, the technical features of the disclosure will be further described in detail below in conjunction with the accompanying drawings.

(7) The disclosure provides a personal audio system, in particular to a mechanical structure of the personal audio system. The personal audio system comprises a housing **10**, in which a right front loudspeaker **31**, a left front loudspeaker **32** and a controller **50** are arranged, the controller **50** is used for controlling the loudspeakers, and the right front loudspeaker **31** and the left front loudspeaker **32** are symmetrically arranged. The personal audio system can be mounted in a

headrest of a seat and can also be mounted in a neck pillow to be worn on the shoulders.

(8) At least four embodiments of the disclosure are provided below.

(9) Embodiment one: As shown in FIG. 1, a personal audio system comprises a housing **10** and a controller **50**. A left front loudspeaker **32** and a right front loudspeaker **31** are arranged in the housing **10** and correspond to the left ear and the right ear respectively. To facilitate description, only design of the right channel is introduced below because the left channel and the right channel has the same design. A right front conduction tube **43** is arranged on one side of the right front loudspeaker **31**, and the right front conduction tube **43** is tightly connected and sealed to the right front loudspeaker **31**. The other end of the right front conduction tube **43** is close to the right ear by a distance a from the ear canal opening, which meets the range condition of $30\text{ mm} \leq a \leq 200\text{ mm}$, and on the premise of not affecting normal use of a user, the smaller the distance a is, the better the tone is. The other side of the right front loudspeaker **31** is tightly connected and sealed to the right rear conduction tube **41**, the other end of the right rear conduction tube **41** is far away from the right ear, and the shortest propagation distance b of sound waves from the right ear meets the condition of $b/a > 2$. If $b/a > 8$, the sound effect heard by the right ear is almost the optimal.

(10) A distance from a distal point M to the user is usually greater than 500 mm , so that a difference c between distances from the tail ends of the right front conduction tube **43** and the right rear conduction tube **41** to the point M meets $c/d < 1$. Since the sound waves emitted from the front side and the back side of the right front loudspeaker **31** are the same in amplitude and opposite in phase, when the sound waves are propagated to the point M , the sound waves offset each other, thereby weakening the sound heard by the user at the point M . In order to achieve the best sound reduction effect, one of preferable manners is that the right front conduction tube **43** and the right rear conduction tube **41** have similar sectional areas and lengths, with an error not exceeding 10% .

(11) If the right front conduction tube **43** and the right rear conduction tube **41** cannot be designed to be similar in size, accurate sound reduction can be achieved by measuring sound wave characteristics at conduction tube openings. After the design of the right front conduction tube **43** is finalized, the sound wave at the opening of the right rear conduction tube **41** is measured. If the sound wave amplitude of the opening of the right rear conduction tube **41** is greater than that of the opening of the right front conduction tube **43**, the sectional area of the right rear conduction tube **41** may be decreased, or the length of the right rear conduction tube **41** may be increased. On the contrary, the sectional area of the right rear conduction tube **41** is increased or the length of the right rear conduction tube **41** is decreased, and this adjustment is performed until a difference between sound wave frequency response curves at the two conduction tube openings is not greater than 6 dB after the two curves are subjected to $\frac{1}{3}$ octave smoothing.

(12) Embodiment two: The right front conduction tube **43** and the right rear conduction tube **41** of the embodiment described above may both be sized according to requirements. When no front conduction tube is arranged due to shortening, the embodiment is illustrated as shown in FIG. 2. An active sound reduction apparatus comprises a housing **10**, a controller **50**, a right front loudspeaker **31** and a left front loudspeaker **32**. To reduce the influence on the user, the right front loudspeaker **31** and the left front loudspeaker **32** are both located behind or below the right and left ears, and the distance a from the ear canal opening of the user meets the condition of $30\text{ mm} \leq a \leq 200\text{ mm}$. When the distance is less than 30 mm , the apparatus is likely to be in contact with the human ears, causing discomfort. When the distance is greater than 200 mm , the stability of a sound field is poor, and the effect is also poor.

(13) The radiation axes of the two loudspeakers point to the right and left ears, so that an included angle θ is formed and meets the condition of $30^\circ < \theta < 170^\circ$. Since the right front loudspeaker **31** and the left front loudspeaker **32** and their related accessories are symmetrical bilaterally, only the design of the right channel will be described next. The right front loudspeaker **31** is connected and sealed to one end of the right rear conduction tube **41**, and an opening at the other end of the right rear conduction tube **41** is away from the right ear. The shortest propagation distance between the

distal opening of the right rear conduction tube **41** and the sound wave to the right ear is represented by b , and if $b/a > 2$, a good bass effect can be obtained; and if $b/a > 8$, the optimal effect can be obtained.

(14) For the user at the distal point M , a distance c between the right front loudspeaker **31** and the point M is almost the same as a distance d between the opening of the right rear conduction tube **41** and the point M , so that the sound waves radiated from the front side and the back side of the right front loudspeaker **31** still meet the offset condition at the point M . In order to achieve the optimal sound reduction effect, it is required to make adjustment according to the sound wave characteristics at the openings of the right front loudspeaker **31** and the right rear conduction tube **41**. Generally, the sound wave radiated by the right front loudspeaker **31** is greater, and it is required to adjust the radiation area of the opening of the right front loudspeaker **31**. By decreasing the area of the opening and increasing radiation damping, the sensitivity of the sound wave directly radiated by the right front loudspeaker **31** can be reduced, and the optimal offset effect that the difference between the sound wave frequency response curves is not greater than 6 dB after the two curves are subjected to $\frac{1}{3}$ octave smoothing can be obtained.

(15) Embodiment three: As shown in FIG. 3, a right rear loudspeaker **33** and a left rear loudspeaker **34** are added on the basis of the embodiment of FIG. 2. The sound waves radiated outwards by the rear loudspeaker and the front loudspeaker are the same in amplitude and opposite in phase.

(16) Since the right rear conduction tube **41** and the left rear conduction tube **42** are additionally arranged on the back of the right front loudspeaker **31** and the back of the left front loudspeaker **32** respectively, the sound waves at the openings of the rear conduction tubes (**41** and **42**) are less than the sound waves radiated by the front loudspeakers (**31** and **32**), and therefore the rear loudspeakers (**33** and **34**) are additionally arranged. A difference value is calculated by measuring the sound waves radiated directly by the front loudspeakers (**31** and **32**) and the sound waves radiated from the openings of the rear conduction tubes (**41** and **42**). The response characteristics of the rear loudspeakers (**33** and **34**) are then controlled by the controller **50**, so that the sound waves replayed by the rear loudspeakers (**33** and **34**) are just equal to the difference value, thereby achieving a better sound reduction effect.

(17) Embodiment four: In the embodiment of FIG. 4, the rear conduction tubes (**41** and **42**) are removed on the basis of the embodiment of FIG. 3, saving the cost. The sound waves radiated by the front loudspeakers (**31** and **32**) toward a surrounding space are offset only by the rear loudspeakers (**33** and **34**). This design can enable most of the functions of FIG. 3, but the bass effect is slightly poor.

(18) The above embodiments are only preferred embodiments of the disclosure and are not intended to limit the concept and scope of the disclosure, and without departing from the gist and scope of the disclosure, various modifications and improvements made on the technical scheme of the disclosure by a person of ordinary skill in the art shall fall within the claimed protection scope of the disclosure.

Claims

1. A personal audio system, comprising: a housing in which a right front loudspeaker, a left front loudspeaker and a controller are all arranged; wherein a right rear conduction tube is arranged on the right front loudspeaker, an opening at one end of the right rear conduction tube is tightly connected and sealed to one side of the right front loudspeaker, and an opening at the other end of the right rear conduction tube is away from a right ear of a user; a left rear conduction tube is arranged on the left front loudspeaker, an opening at one end of the left rear conduction tube is tightly connected and sealed to one side of the left front loudspeaker, and an opening at the other end of the left rear conduction tube is away from a left ear of the user; wherein a right front conduction tube is arranged on the right front loudspeaker, an opening at one end of the right front

conduction tube is tightly connected and sealed to the other side of the right front loudspeaker, and an opening at the other end of the right front conduction tube is close to the right ear of the user; a left front conduction tube is arranged on the left front loudspeaker, an opening at one end of the left front conduction tube is tightly connected and sealed to one side of the left front loudspeaker, and an opening at the other end of the left front conduction tube is close to the left ear of the user.

2. The personal audio system according to claim 1, wherein the right front loudspeaker and the left front loudspeaker are the same in structure and are symmetrically arranged, the left front conduction tube and the right front conduction tube are the same in structure and are symmetrically arranged, a distance a between the right front conduction tube and an ear canal opening ranges from 30 mm to 200 mm, a distance by which a sound wave is radiated by air from a distal end of the right rear guide tube to the right ear is represented by b , and $b/a > 2$.

3. The personal audio system according to claim 2, wherein $b/a > 8$; an error between sectional areas of the right front conduction tube and the right rear conduction tube does not exceed 10%, and an error between lengths of the right front conduction tube and the right rear conduction tube does not exceed 10%.

4. The personal audio system according to claim 1, wherein the right front loudspeaker and the left front loudspeaker are the same in structure and are symmetrically arranged, the right rear conduction tube and the left rear conduction tube are the same in structure and are symmetrically arranged, and radiation axes of the right front loudspeaker and the left front loudspeaker respectively point to two ears of the user; an included angle θ formed by the two axes ranges from 30° to 170° , a distance a by which a sound wave is radiated from a front side of the right front loudspeaker to the right ear pointed by the right front loudspeaker ranges from 30 mm to 200 mm, a distance by which a sound wave is radiated by air from a distal end of the right rear conduction tube to the right ear is represented by b , and $b/a > 2$.

5. The personal audio system according to claim 4, wherein $b/a > 8$.

6. The personal audio system according to claim 1, wherein the right front loudspeaker and the left front loudspeaker are the same in structure and are symmetrically arranged, a right rear loudspeaker and a left rear loudspeaker are symmetrically arranged in the housing, the right rear loudspeaker is reversely arranged behind the right front loudspeaker, and the left rear loudspeaker is reversely arranged behind the left front loudspeaker.
